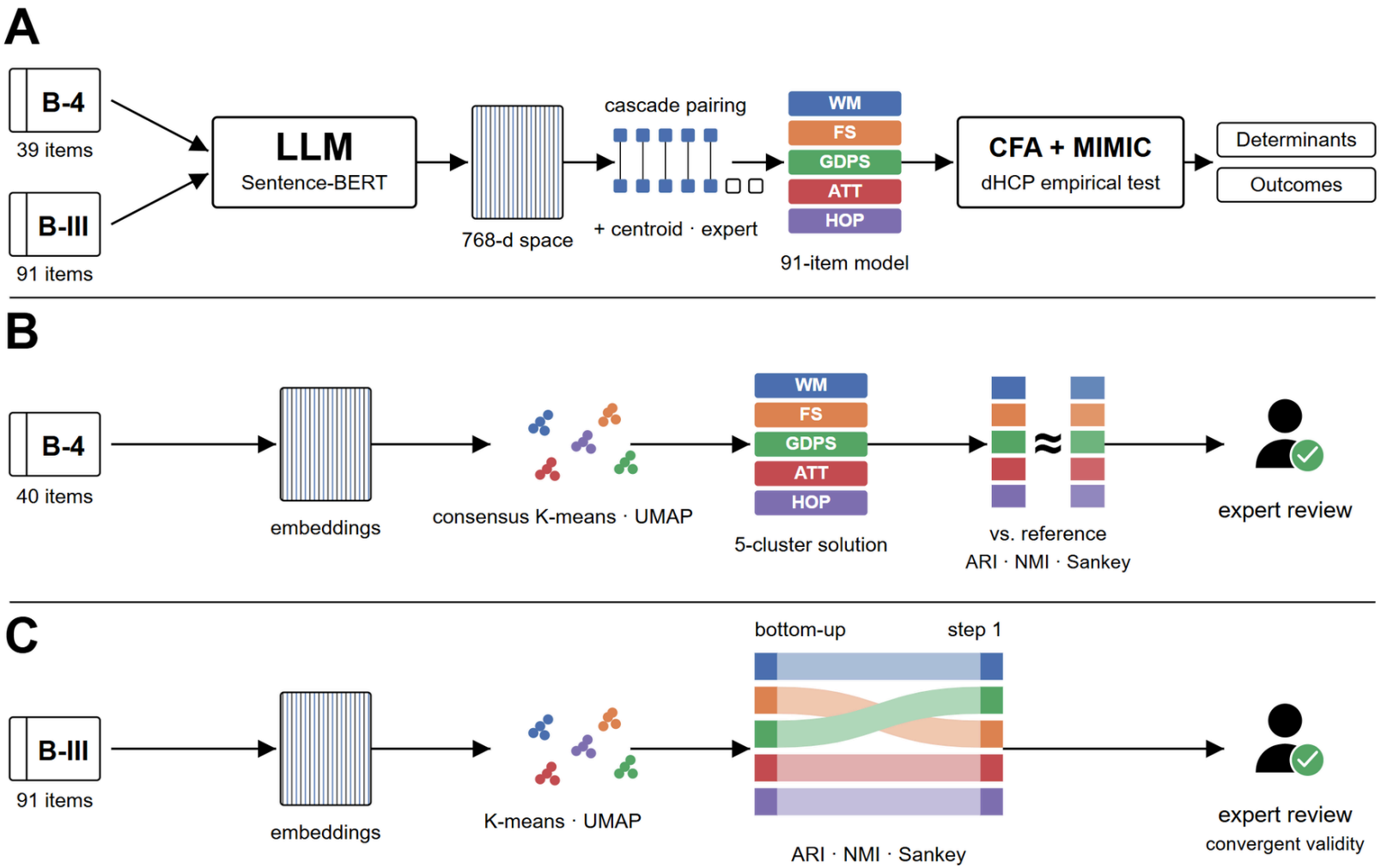
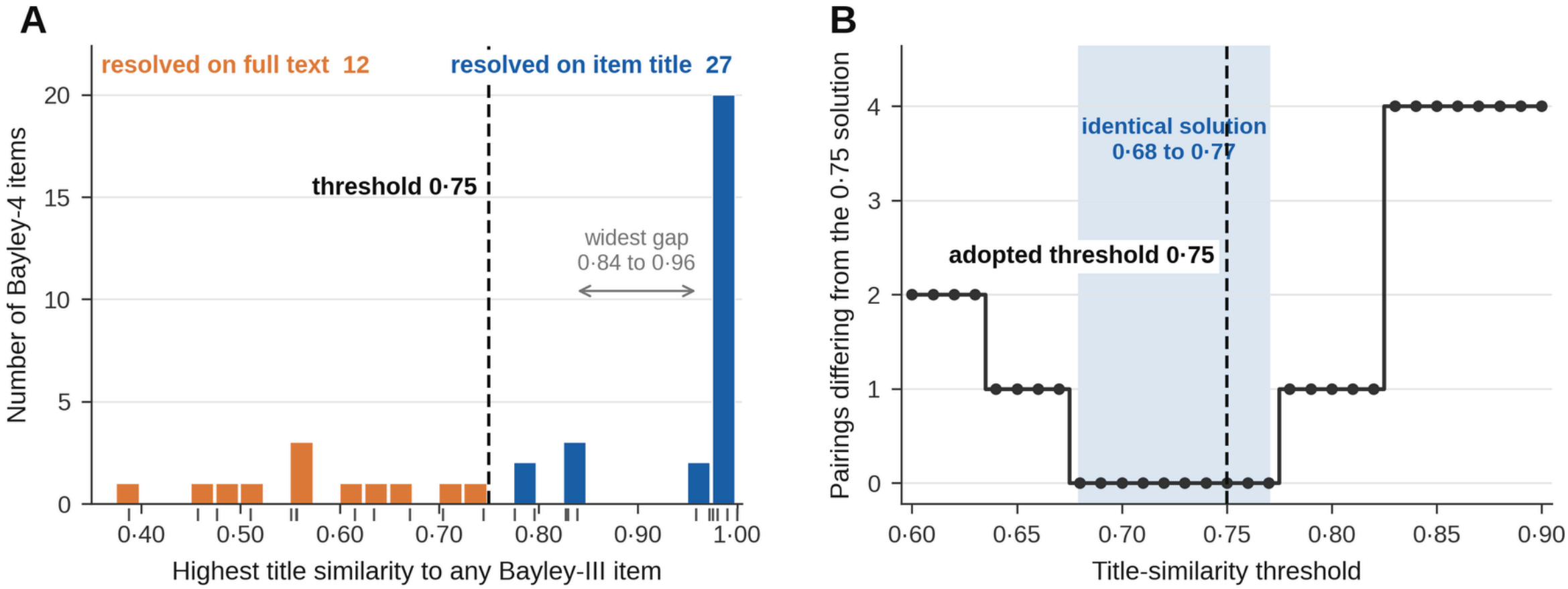
The pipeline, in one figure

Both item sets are embedded by the same encoder into a 768-d space. Bayley-III items are paired with their closest Bayley-4 item; the rest go to domain centroids, then expert review. The 91-item model is tested with CFA and MIMIC in 739 dHCP infants. Rows B and C are unsupervised convergence checks (consensus K-means, ARI, NMI, Sankey).



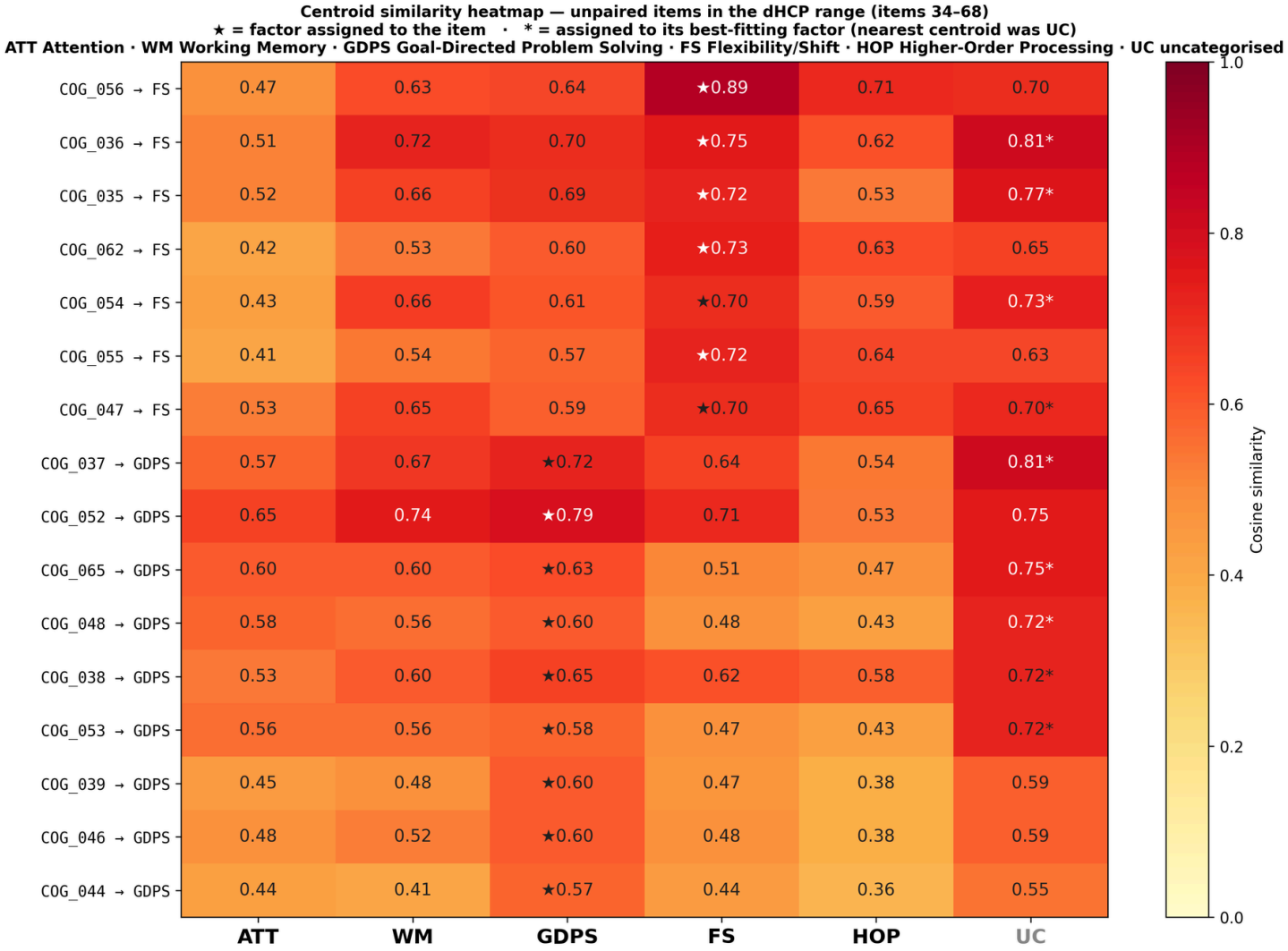
Pairing each Bayley-III item with its Bayley-4 counterpart

Cascade: titles embedded and compared first; a best match above 0.75 is title-locked (27 items), the rest are resolved on full item text (12). Any threshold from 0.68 to 0.77 yields the identical pairing.



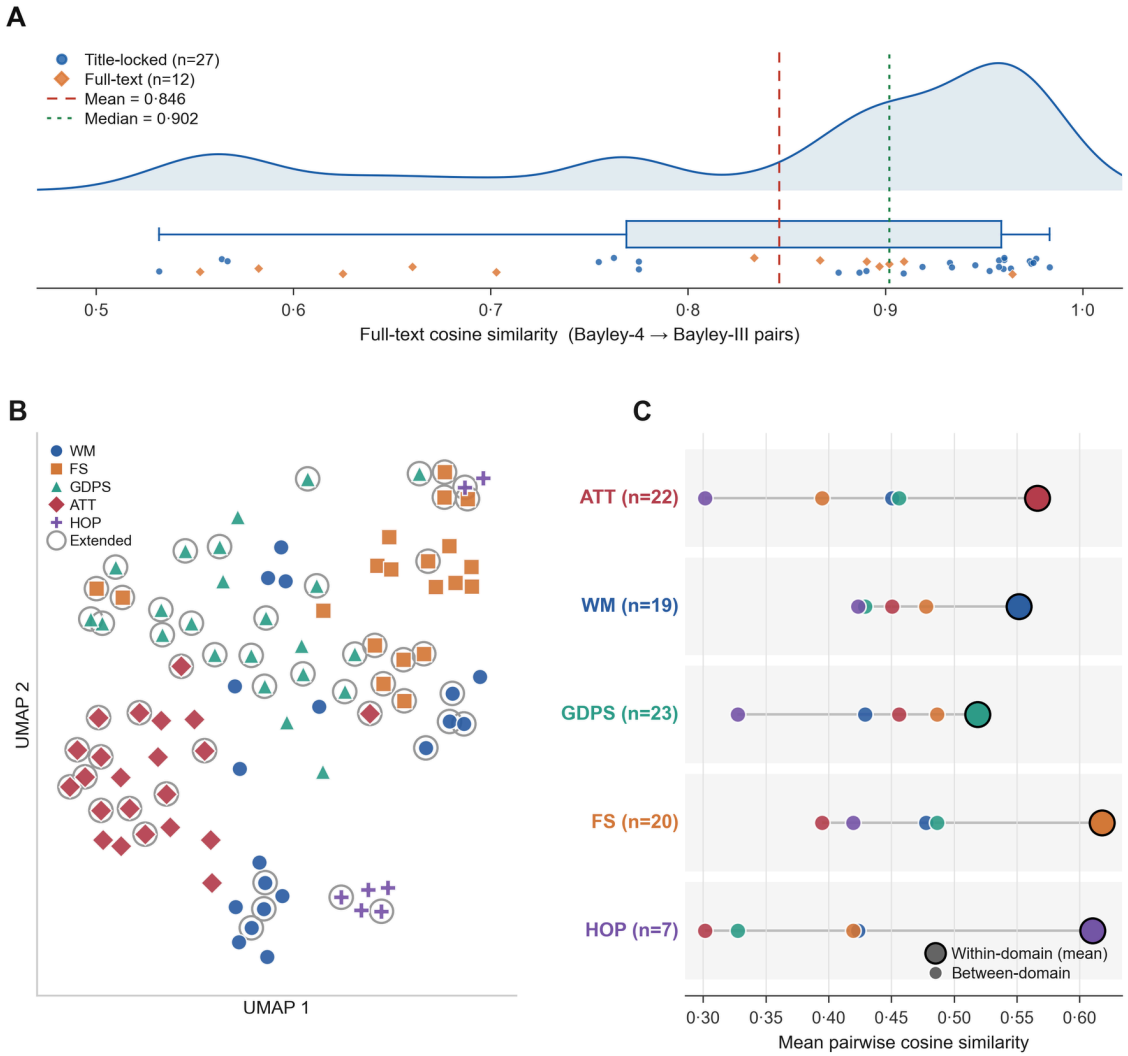
Items with no counterpart: nearest domain centroid

52 Bayley-III items have no Bayley-4 counterpart. Each domain gets a centroid (mean embedding of its paired items); the item goes to the nearest. Star = assigned domain. A sixth, uncategorised cluster flags items for co-author review when it wins by more than 0.10.



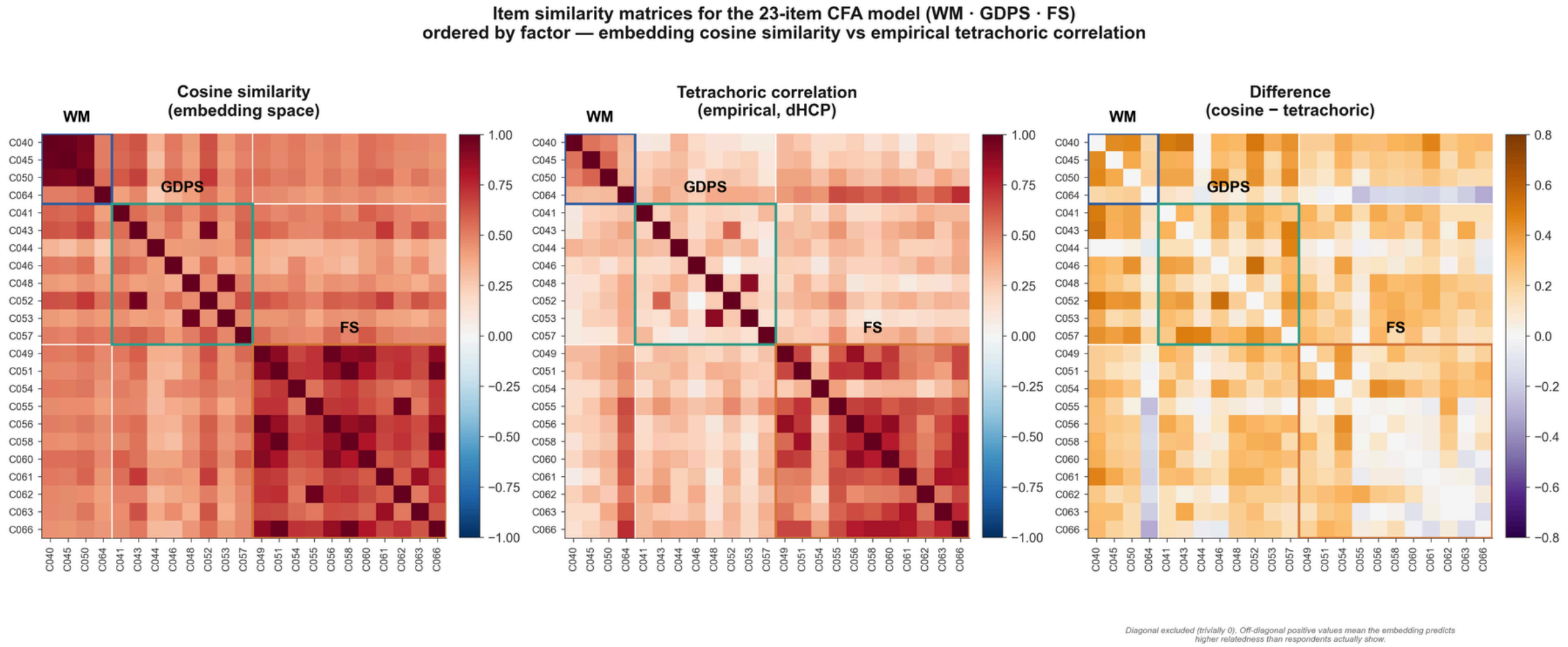
The translated structure in embedding space

A: full-text similarity of the 39 pairs, mean 0.846, median 0.902. B: UMAP of all 91 items by domain; attention and flexibility/shift form regions, working memory and problem solving interleave. C: within-domain similarity exceeds between-domain similarity for every domain.



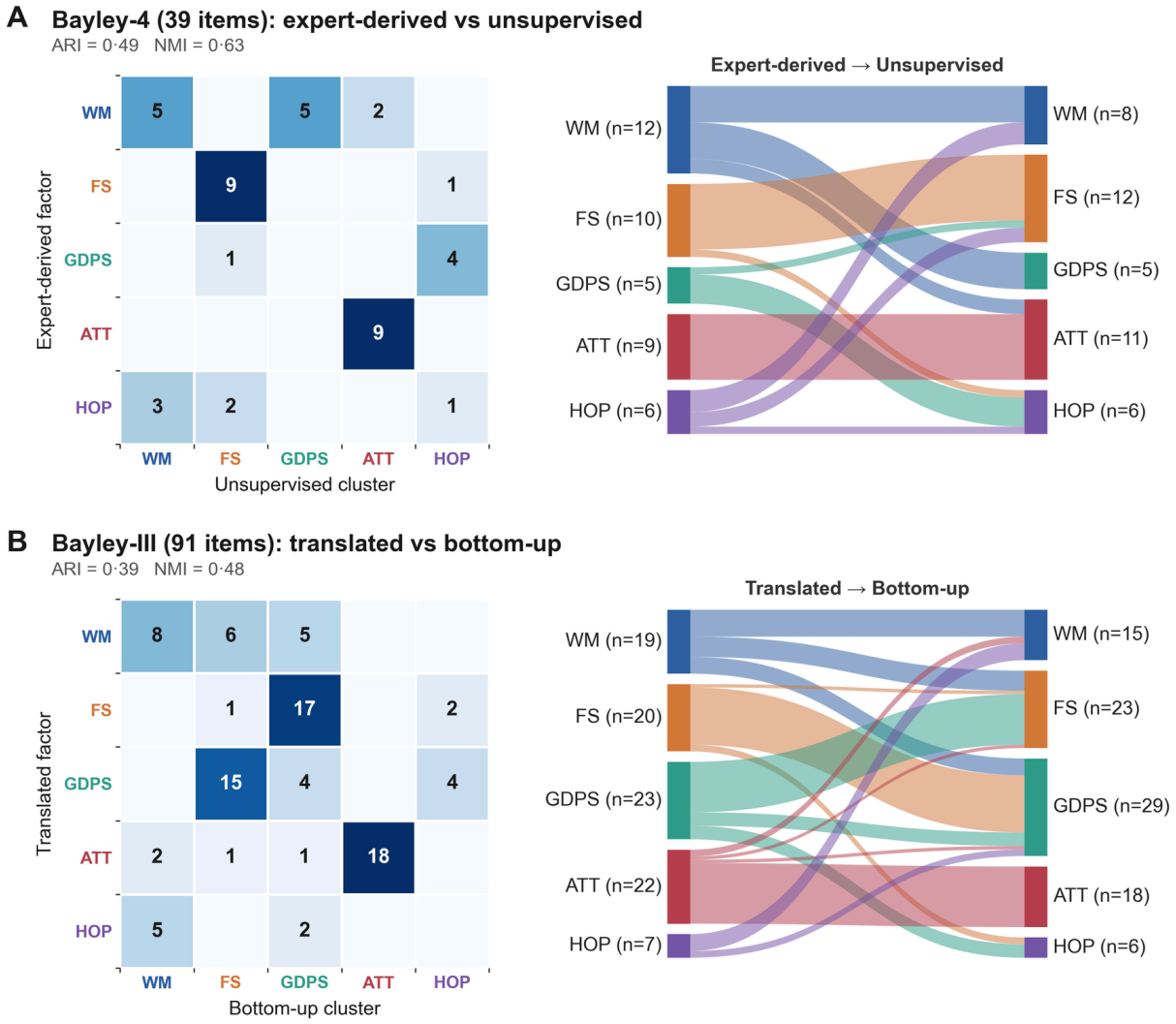
Does embedding similarity predict how infants respond?

Exercise 2, section 4, on real data: cosine similarity (left) versus tetrachoric correlations between pass/fail responses (middle) for the 23 CFA items. The block structure is shared; the difference (right) is almost entirely positive: text similarity is an upper bound on empirical similarity.



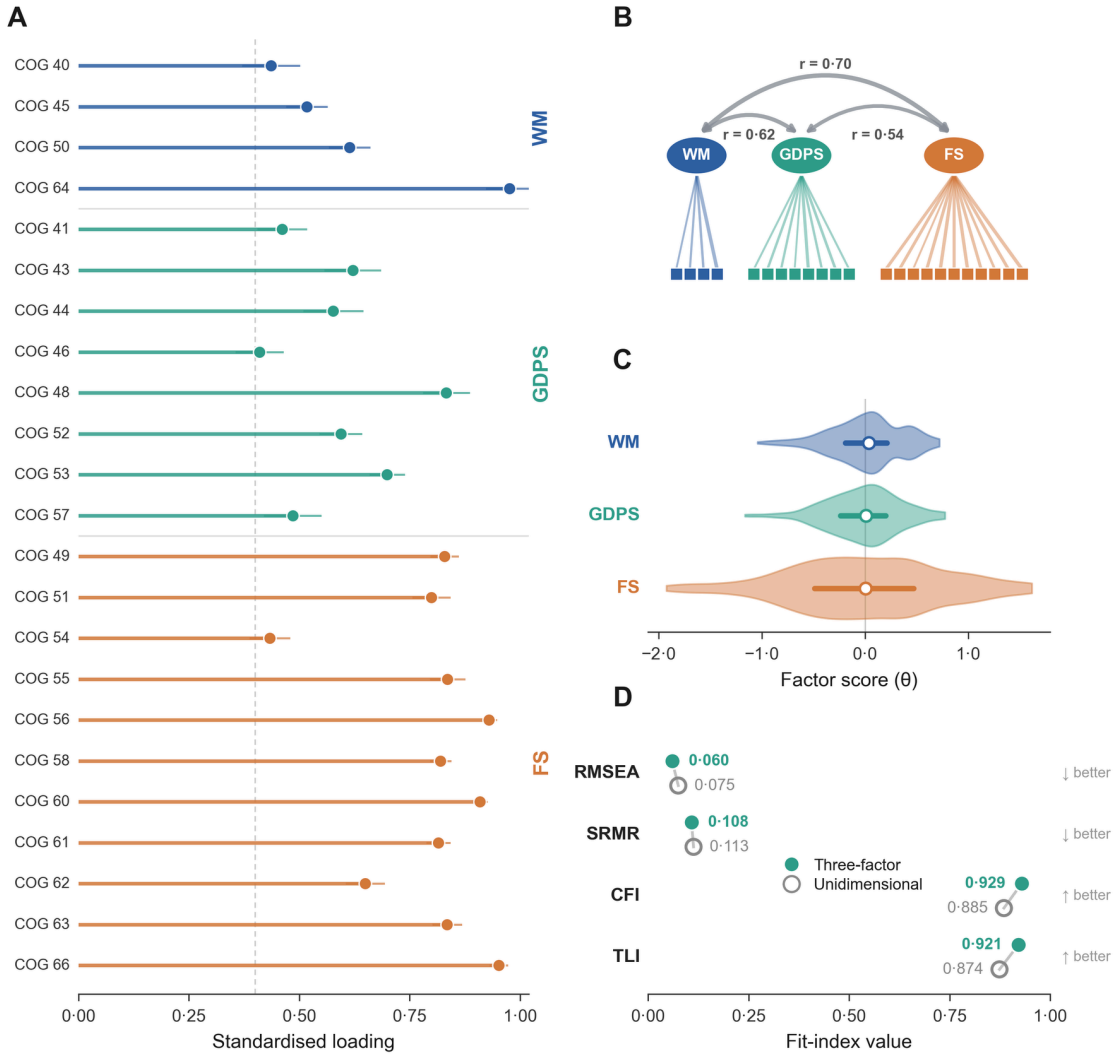
Is it the embedding talking? Unsupervised convergence check

Consensus K-means with no labels. Bayley-4: clusters vs expert structure ARI 0.49, NMI 0.63. Bayley-III: clusters vs translated structure ARI 0.39, NMI 0.48. Attention and flexibility/shift come back intact; working memory splits toward problem solving. A convergence check, not a replacement.



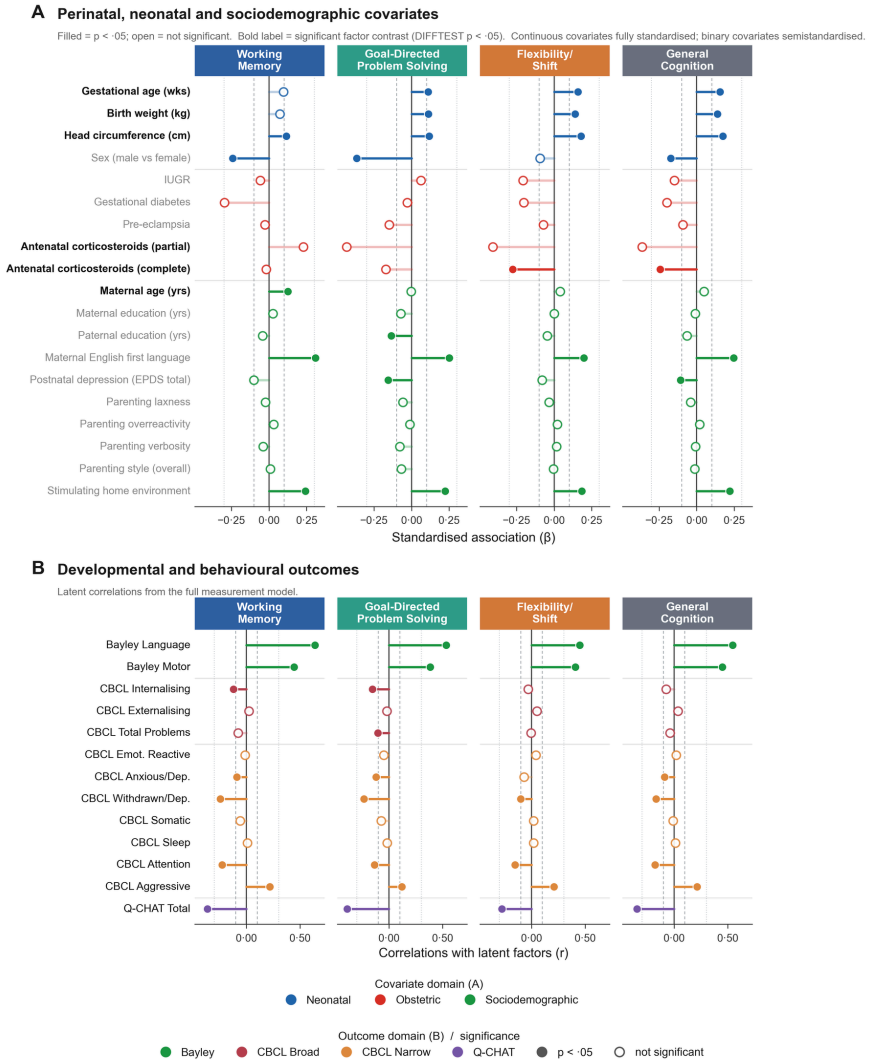
Confirmatory test in 739 infants

Three testable domains (one attention and one higher-order item in range). All 23 loadings above 0.40; factor correlations 0.54 to 0.70; three-factor beats unidimensional on every index: RMSEA 0.060 vs 0.075, CFI 0.929 vs 0.885, TLI 0.921 vs 0.874.



What the domains reveal that the composite hides

Gestational age, birth weight and head circumference relate to problem solving and flexibility/shift, not working memory. The sex difference is confined to working memory and problem solving. Complete antenatal corticosteroids act on flexibility/shift. Later internalising problems track problem solving. The composite dilutes or misses each.



Back to the seminar

Items are text; proximity is similarity; the structure read from text was then judged by response data.

Two caveats carry over from the exercises: general-purpose models capture what an item is about, not how it is keyed; a model fine-tuned on the items you evaluate needs a held-out test.

Book, exercises and solutions: https://leonardozaggia.github.io/PONS_material/
Paper: Zaggia, Diederich, Hilgendorff, Heep and Hildebrandt, From Composite to Components, under review.
Questions: leonardo.zaggia@uol.de